

Jawless fish

The first vertebrates were fish, but they were very different from today's fish. The early fish couldn't bite, since jaws were yet to evolve. Instead, they fed by sucking or scraping. With few or no fins, they swam by wagging their tails like tadpoles. They had no internal bones, but some had wide, bony shields covering their heads—protection from predators such as giant sea scorpions.

Drepanaspis

DREP-an-ASP-iss

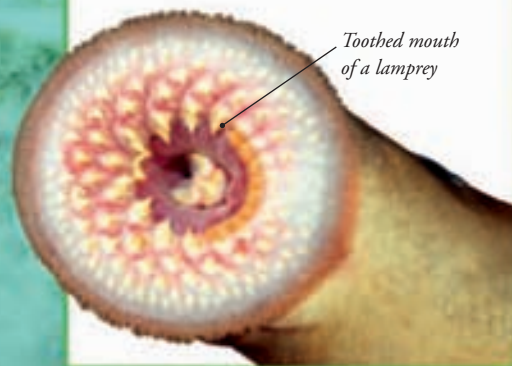


- **When** 410 million years ago (Early Devonian)
- **Fossil location** Europe
- **Habitat** Ocean floor
- **Length** 14 in (35 cm)

With its flat, paddle-shaped head and narrow body, *Drepanaspis* was a strangely shaped fish. It hunted for food near the bottom of Devonian seas. *Drepanaspis*'s feeding methods are a mystery, since its jawless mouth faced upward instead of downward, a curious feature that would have made it difficult to scoop in food. Like many other jawless fish, *Drepanaspis* had bony armor to protect it from attack.

LIVING RELATIVE

Two groups of jawless fish still exist today: hagfish and lampreys. Both are eel-shaped creatures with no bones, scales, or fins. Hagfish feed on worms or dead sea animals. Some lampreys are parasites – they use their circular, jawless mouths to latch on to fish so they can feed on their blood.



Toothed mouth of a lamprey